

PROJECT REPORT

CUSTOMER RETENTION & CHURN ANALYSIS & DASHBOARD – TELCO CUSTOMER CHURN DATASET

Internship

Data Science and Analytics Internship

Organization

Future Interns



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Summary

- This project analyzes customer churn and retention using the Telco Customer Churn Dataset. The objective is to identify key factors influencing customer churn and understand customer lifetime behavior. Using Excel and Python, the data was cleaned, analyzed, and visualized through an interactive dashboard. Key metrics such as churn rate, retention rate, and average tenure were examined alongside factors like contract type, payment method, and service usage. The analysis highlights that churn is higher among short-term customers and during the early stages of the customer lifecycle. The project provides actionable insights to improve retention and support data-driven decision-making.
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Problem Statement

- Customer churn is a major challenge for subscription-based businesses, directly impacting revenue and growth. Organizations need to understand why customers leave and what factors influence retention. This project aims to analyze customer data to identify churn patterns, key drivers of customer behavior, and customer lifetime trends, in order to provide actionable insights that help improve retention and support data-driven business decisions.

Project Objectives

- To analyze customer churn and retention patterns using the Telco Customer Churn Dataset
- To identify key factors influencing customer churn, such as contract type, payment method, and service usage
- To study customer lifetime behavior by analyzing tenure and grouping customers into meaningful segments
- To understand at which stage customers are most likely to churn
- To develop an interactive dashboard for clear visualization of churn patterns and retention drivers
- To enable dynamic data exploration using slicers and filters
- To generate actionable insights that help reduce customer churn
- To provide data-driven recommendations for improving customer retention and business performance

Data Description

The dataset used for this project is the Telco Customer Churn Dataset, which contains information about customer demographics, services, and subscription behavior.

- The dataset consists of approximately **7,000+** customer records
- Each record represents an individual customer
- The dataset includes **21 columns** with a mix of categorical and numerical variables

Key Feature

- tenure – Customer lifetime
- Contract – Key churn driver
- PaymentMethod – Customer payment behavior
- InternetService – Type of service used
- MonthlyCharges – Pricing factor
- Churn – Target variable (churn status)
- SeniorCitizen – Customer segmentation

DATA PREPROCESSING

- **Data Loading** – The dataset was imported into Python using Pandas for analysis.
- **Data Inspection** – The structure, data types, and completeness of the dataset were examined.
- **Handling Data Types** – The TotalCharges column was converted to numeric format for accurate analysis.
- **Handling Missing Values** – Missing and inconsistent values were identified and handled appropriately.
- **Encoding Target Variable** – The Churn column was encoded as 0 (Retained) and 1 (Churned).
- **Feature Engineering** – A new column, Tenure Group, was created to analyze customer lifetime segments.
- **Data Validation** – The dataset was verified to ensure accuracy and readiness for analysis.

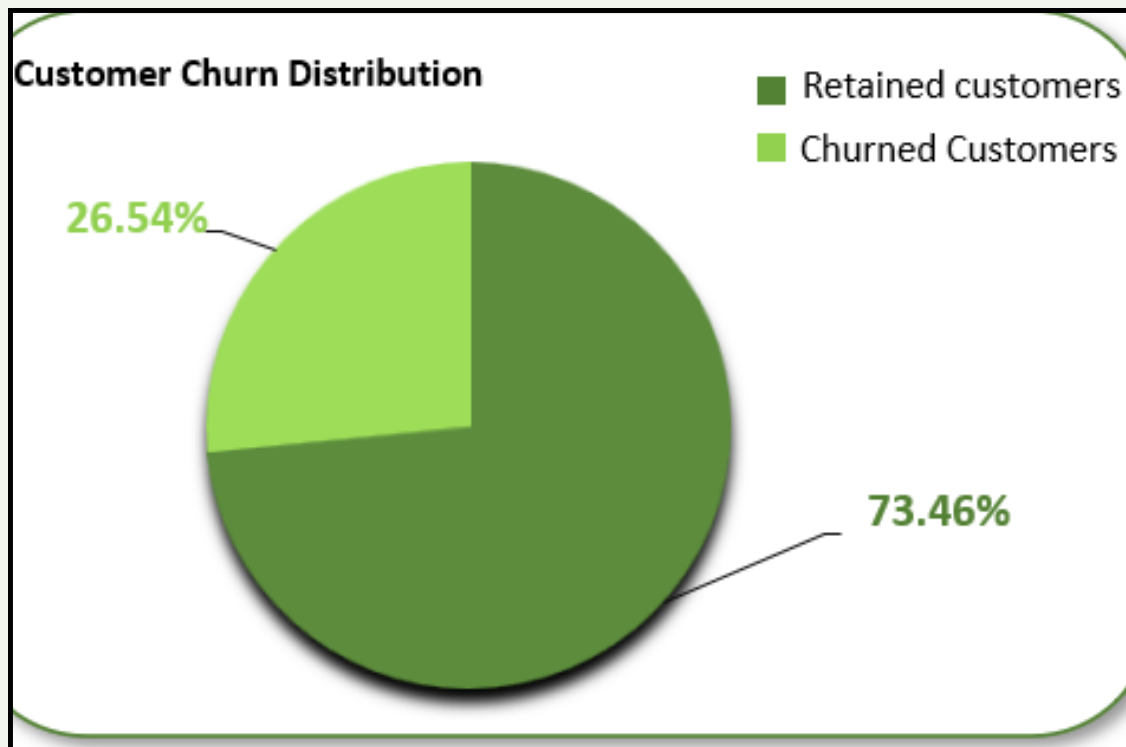
Derived Feature

- **Tenure Group** – Created to categorize customers into:
 - 0–1 Year
 - 1–2 Years
 - 2+ Years

This feature helps in analyzing customer lifetime trends more effectively.

Exploratory Data Analysis (EDA)

1. Customer Churn Distribution

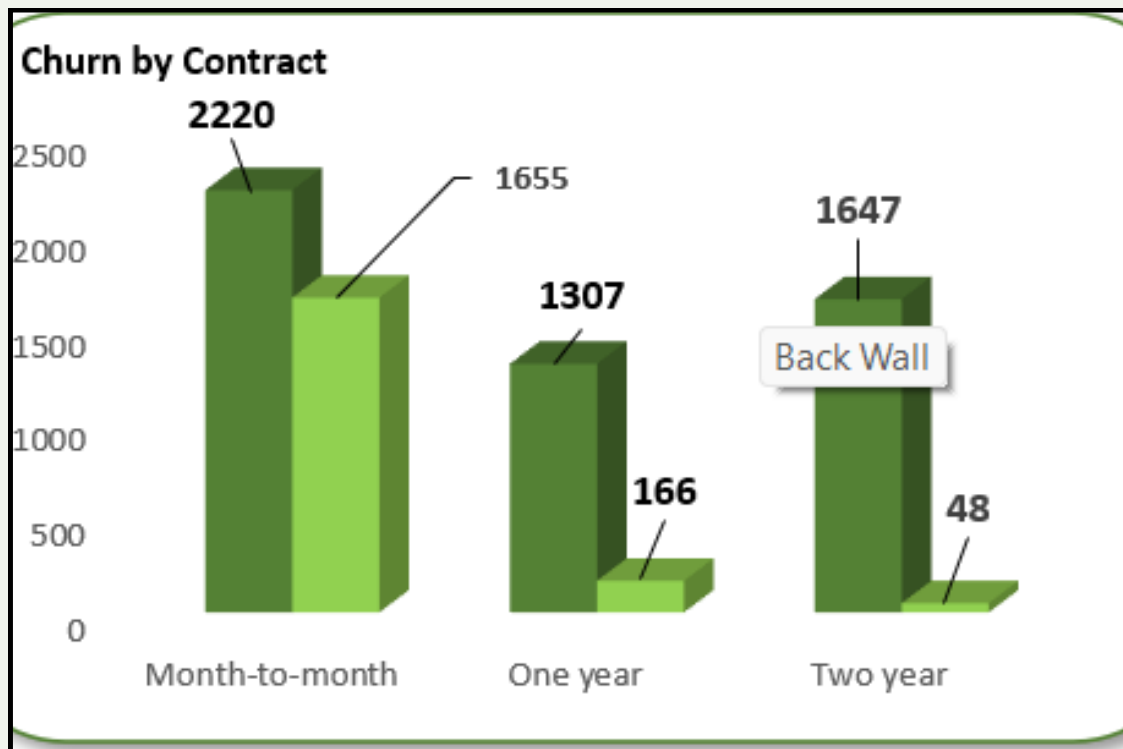


This chart illustrates the overall distribution of customers based on churn status. It is observed that approximately **26.6% of customers have churned**, while the remaining **73.4% are retained**. Although the majority of customers continue using the service, the churn percentage is significant and indicates a potential risk to long-term revenue. This highlights the need for targeted retention strategies to reduce customer attrition.

Insight:

Customers who churn represent over one-fourth of the total customer base, indicating a significant retention challenge that requires immediate business attention.

2.Churn by Contract Type

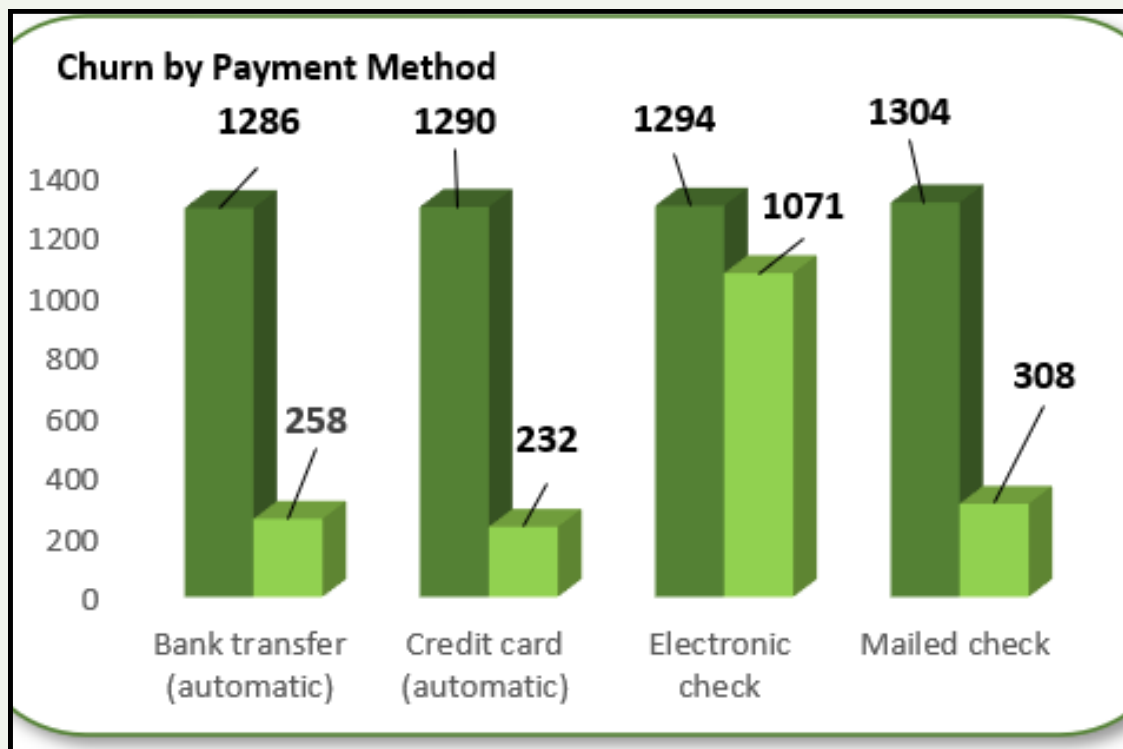


- The analysis shows that customers with **month-to-month** contracts have the highest churn compared to those with one-**year and two-year contracts**. This suggests that customers without long-term commitment are more likely to leave the service. In contrast, long-term contracts provide stability and improve customer retention. This insight indicates that encouraging customers to opt for longer-duration plans can significantly reduce churn.

Insight:

Customers without long-term commitment are more likely to leave, suggesting that promoting **long-term subscription plans** can effectively reduce churn and improve retention.

3.Churn by Payment Method

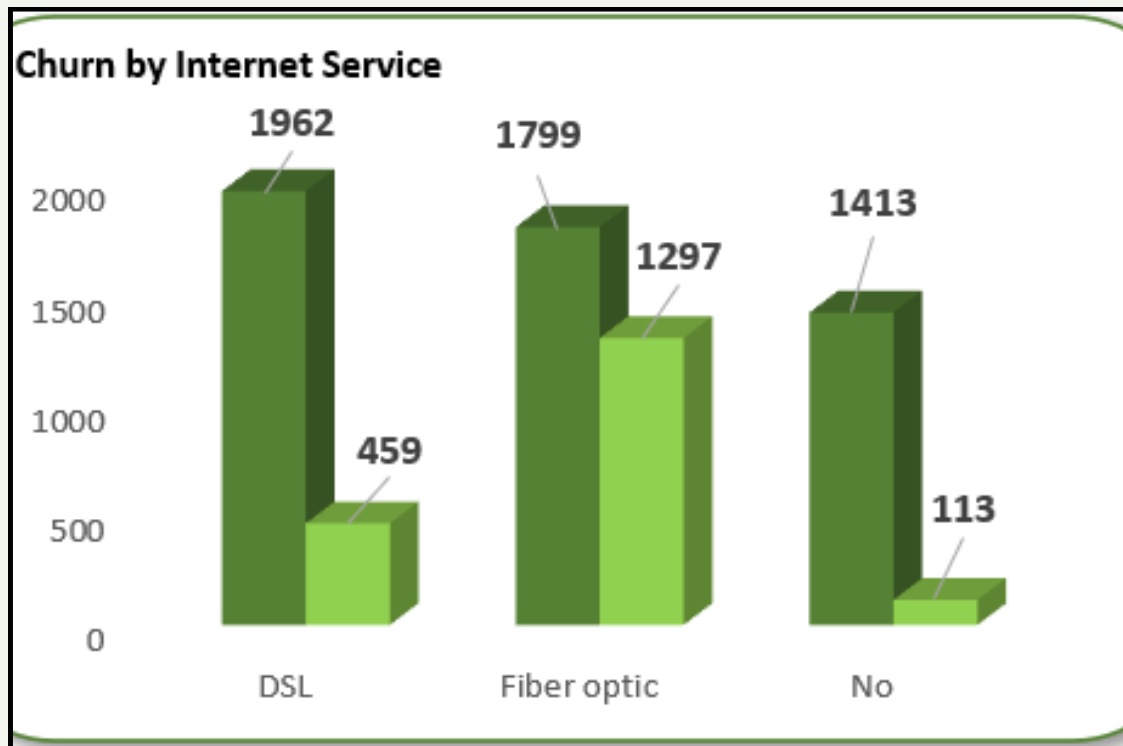


■ Customers using **electronic check** as their payment method exhibit a higher churn rate compared to those using **automatic payment methods** such as bank transfers or credit cards. This may indicate that manual or less convenient payment options contribute to dissatisfaction or discontinuity in service usage. Promoting seamless and automated payment methods can help improve customer retention.

Insight:

Customers using manual or less convenient payment methods are more likely to churn. Encouraging customers to switch to **automatic payment methods** can help improve retention and reduce churn.

4.Churn by Internet Service

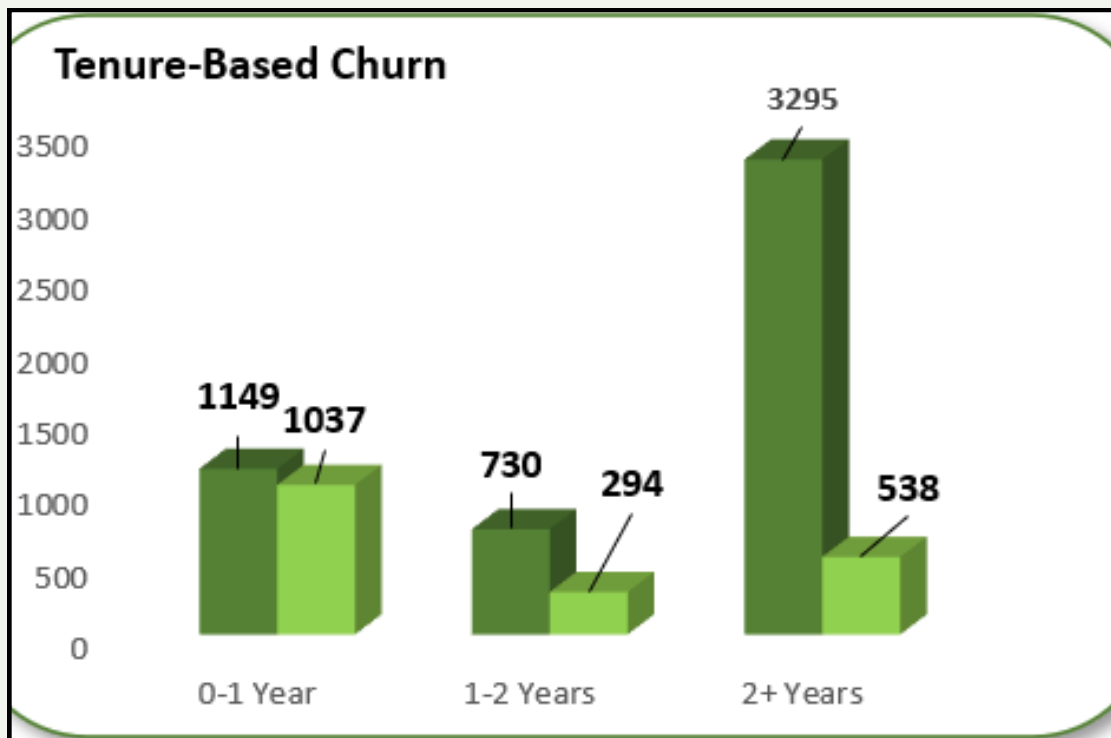


- The chart reveals that customers subscribed to **fiber optic services** experience higher churn compared to DSL and non-internet users. This could be due to factors such as higher pricing, service quality issues, or unmet customer expectations. This insight suggests that businesses should focus on improving service quality or pricing strategies for fiber optic users to reduce churn

Insight:

Higher churn among fiber optic users may indicate issues related to **pricing, service quality, or customer expectations**. Improving service experience or revising pricing strategies for fiber optic customers can help reduce churn.

5. Customer Lifetime Analysis (Tenure-Based Churn)



- The analysis of tenure groups shows that churn is highest among customers in the **0-1 year category**, while customers with longer tenure (especially **2+ years**) exhibit strong retention. This indicates that customers are more likely to leave during the early stages of their subscription. It highlights the importance of improving onboarding experience, customer engagement, and early support to reduce churn during the initial phase of the customer lifecycle.

Insight:

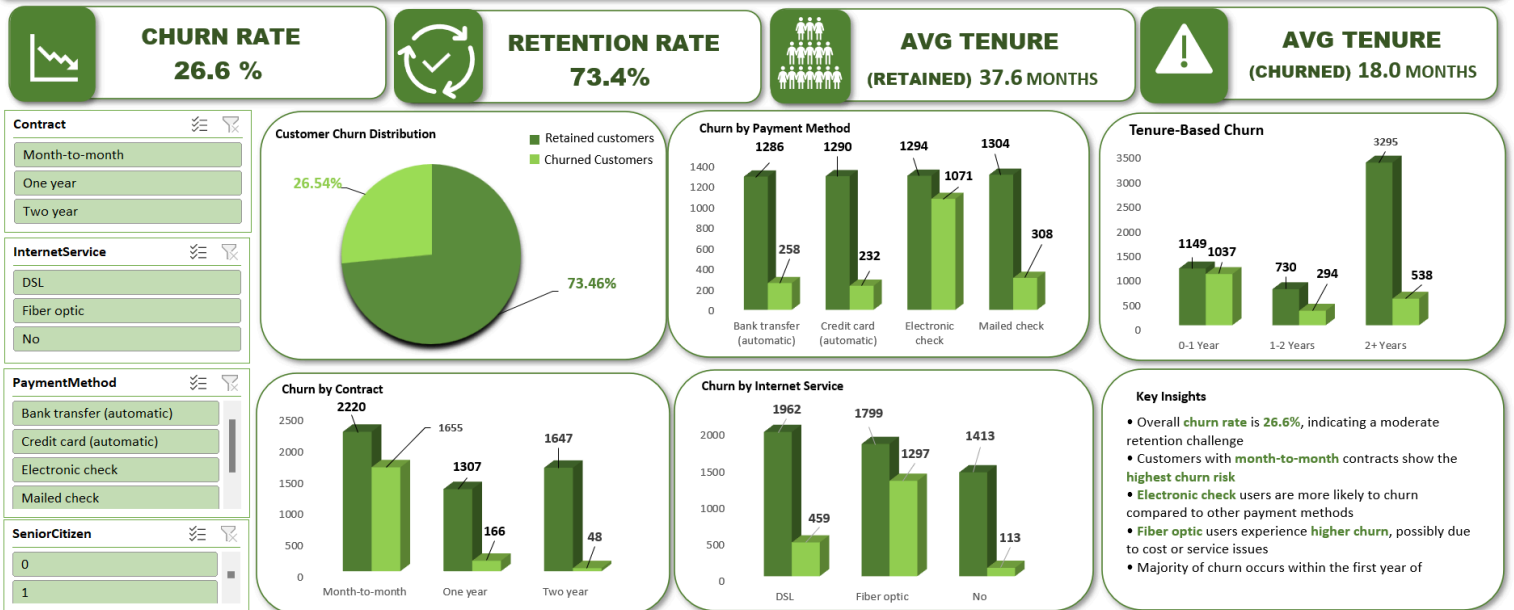
Customer churn is highest during the **initial stages of the customer lifecycle**. Improving onboarding experience, early customer engagement, and support services can help reduce early churn and increase long-term retention.

Dashboard

An interactive dashboard was developed using Excel to visualize customer churn and retention patterns effectively. The dashboard integrates multiple charts and key performance indicators (KPIs) to provide a comprehensive view of customer behavior.

It includes metrics such as **churn rate**, **retention rate**, and **average tenure**, along with visualizations for contract type, payment method, internet service, and customer lifetime analysis. Interactive slicers enable dynamic filtering based on different customer segments, allowing deeper exploration of the data.

CUSTOMER RETENTION & CHURN ANALYSIS DASHBOARD



Insight:

The dashboard provides a consolidated and user-friendly view of key churn drivers, enabling stakeholders to quickly identify high-risk customer segments and make informed, data-driven decisions to improve retention.

Key Insights

- Overall churn rate is 26.6%, indicating a moderate retention challenge
- Customers with month-to-month contracts have the highest churn risk
- Electronic check users are more likely to churn compared to other payment methods
- Fiber optic users show higher churn, possibly due to pricing or service issues
- Majority of churn occurs within the first year of subscription
- Customers with longer tenure (2+ years) show strong loyalty and retention

Business Recommendations

- Encourage customers to opt for long-term contracts to improve retention
- Enhance onboarding experience to reduce early-stage churn
- Promote automatic payment methods for convenience and continuity
- Improve service quality and pricing strategies for fiber optic users
- Focus on early customer engagement and support to build long-term relationships

Conclusion

This project analyzed customer churn using the Telco Customer Churn Dataset and identified key factors influencing customer behavior. The findings show that churn is higher among short-term and early-stage customers. The insights and recommendations provided can help businesses improve retention and make data-driven decisions.

Thankyou